

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models ¹
Student Name	Isha Venkataraman
Register Number	3122247001023
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	30/08/26

1 Objective

- To understand three ensemble learning strategies – Bagging, Boosting and Stacking.
- To implement Bagging and Boosting classifiers on the dataset.
- To build a Stacked Ensemble using multiple base learners with a meta-learner on top.
- To compare the three ensemble methods on accuracy, stability and generalisation.
- To see how each ensemble strategy affects the bias-variance tradeoff.

2 Problem Statement

Given 30 numerical features describing a breast cell nucleus, the task is to classify the sample as Benign or Malignant. Instead of using a single classifier, three different ensemble strategies are built and compared: Bagging (parallel, variance-reducing), Boosting (sequential, bias-reducing) and Stacking (combining different model types through, a meta-learner). The goal is to see which strategy gives the most accurate and stable predictions on this dataset.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30 numerical features
Number of Classes	2 (Benign, Malignant)
Missing Values	None
Train-Test Split	80% – 20%, stratified (455 train, 114 test)

4 Exploratory Data Analysis

Class Distribution

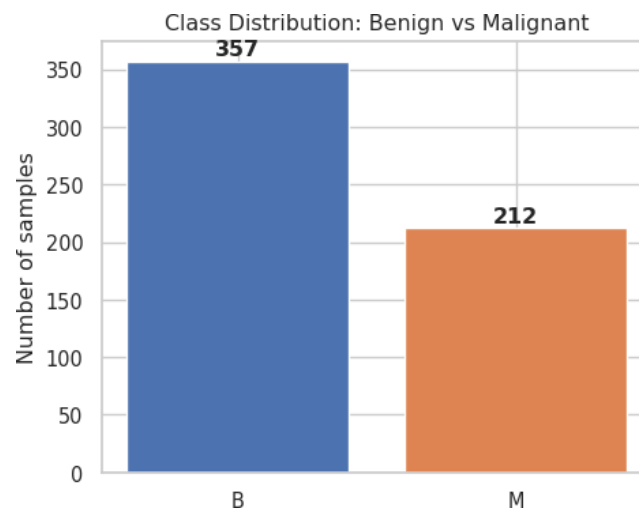


Figure 1: Class distribution: Benign vs Malignant

Inference: Out of 569 samples, 357 are Benign, and 212 are Malignant, roughly a 63:37 split. The dataset is mildly imbalanced but not to a worrying degree. Because of this, accuracy alone is not fully trusted here, and F1-score/recall on the malignant class is tracked more closely, since missing an actual, malignant case is a costlier mistake than a false alarm. Stratified splitting is used, so this ratio stays same in train and test sets.

Correlation Heatmap

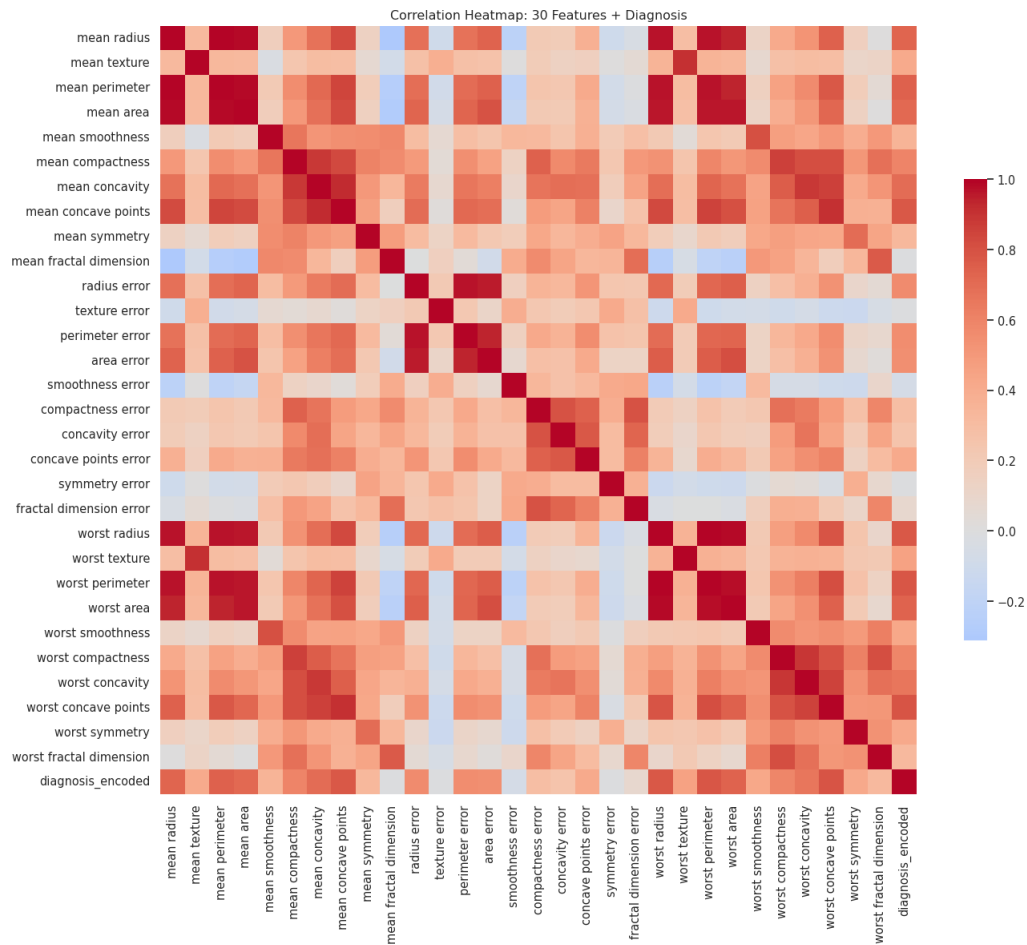


Figure 2: Correlation heatmap of all 30 features with diagnosis

Inference: A lot of the mean, standard-error and worst versions of the same measurement (radius, perimeter, area) are strongly correlated with each other, since they all come from the same cell shape. This kind of overlap between features can be a problem for linear models, but tree-based ensembles used in this experiment are not really affected by it because a tree only ever splits on one feature at a time.

Feature Correlation with Diagnosis

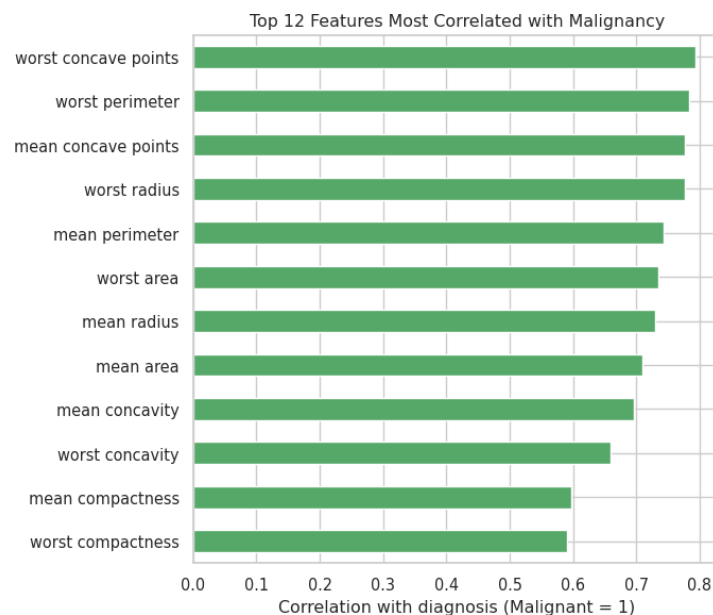


Figure 3: Top 12 features most correlated with malignancy

Inference: The worst (largest observed) values of concave points, perimeter, radius and area come out on top of this list. This matches the general medical idea, that malignant tumours tend to be bigger, and more irregular in shape compared to benign ones. These same features are expected to show up as important, in the ensemble models built later.

5 Data Preprocessing

Missing values: None found, so no imputation was required.

Encoding: Diagnosis label mapped to 1 = Malignant, 0 = Benign.

Feature scaling: StandardScaler applied to all 30 features. This mainly matters for the SVM used inside the Stacked Ensemble later, since SVM is distance-based; the tree-based models (Decision Tree, Bagging, Boosting) do not strictly need scaling, but the same scaled features, are used everywhere to keep the comparison consistent.

Train-test split: 80–20 stratified split, giving 455 training and 114 testing samples, keeping the class ratio the same in both.

6 Machine Learning Algorithms

Bagging

Bagging (Bootstrap Aggregation) trains many copies of the same base estimator, a Decision Tree here, on different bootstrap resamples of the training data, then combines their predictions by averaging/majority voting. Since each tree, is trained on a slightly different sample, their individual mistakes are not perfectly correlated, so combining them

cancels out a good part of the variance.

Advantage: reduces variance and overfitting of the base model.

Limitation: does not help much if the base model itself has high bias.

Boosting (AdaBoost and Gradient Boosting)

Boosting builds models sequentially, where each new model tries to fix the mistakes of the previous ones. AdaBoost increases the weight of misclassified samples so the next learner focuses more on them. Gradient Boosting instead fits each new tree to the residual errors of the ensemble built so far. Because of this sequential correction, boosting mainly targets **bias** rather than variance.

Advantage: can turn a weak learner into a strong one.

Limitation: more sensitive to noisy data and can overfit if run for too many rounds or with a high learning rate.

Stacked Ensemble

Stacking combines predictions from different types of base models : SVM, Gaussian Naive Bayes and a Decision Tree here , and trains a meta-learner (Logistic Regression) on top of their out of fold predictions to learn the best way to combine them.

Advantage: since the base models are different nature, they tend to make different kind of mistakes, and the meta-learner can exploit this diversity.

Limitation: more complex to build and tune, and the meta-learner has very little data which are essentially just the base model outputs to learn from, so it can overfit if not validated carefully.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	1.x
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard CPU (no GPU required)

8 Hyperparameter Selection

All hyperparameters were chosen using GridSearchCV with 5-fold stratified cross-validation, scored on F1 (rather than plain accuracy) since correctly catching malignant cases matters [more here](#).

Model	Best Parameters
Bagging	n_estimators=50, max_samples=0.7, max_features=0.5
AdaBoost	n_estimators=200, learning_rate=0.5
Gradient Boosting	n_estimators=200, learning_rate=0.1, max_depth=2
Stacked Ensemble	SVM + Naive Bayes + Decision Tree → Logistic Regression

9 Results

Metric	Value (Boosting, best overall on F1/Accuracy)
Accuracy	96.49%
Precision	1.00
Recall	0.9048
F1-score	0.95
ROC-AUC	0.9934 (Gradient Boosting), 0.9940 (Stacking)

10 Result Visualization

Performance Comparison

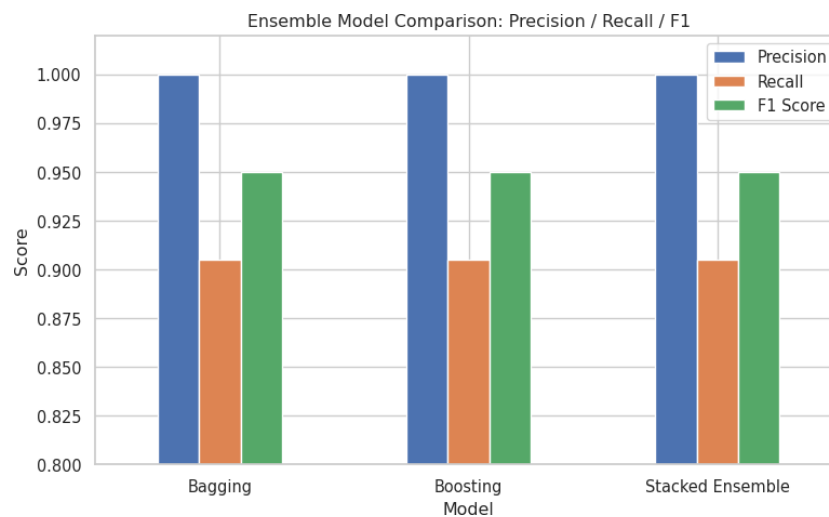


Figure 4: Precision, Recall and F1 comparison across the three ensembles

Inference: All three ensembles score close to each other on this dataset, so the y-axis is zoomed in to make the small differences visible instead of three bars looking identical. Precision is a perfect 1.0 for all three models on the test set, meaning none of them raised a false alarm; recall is where the ensembles differ the most, which directly shows how many actual malignant cases each model caught.

Confusion Matrices

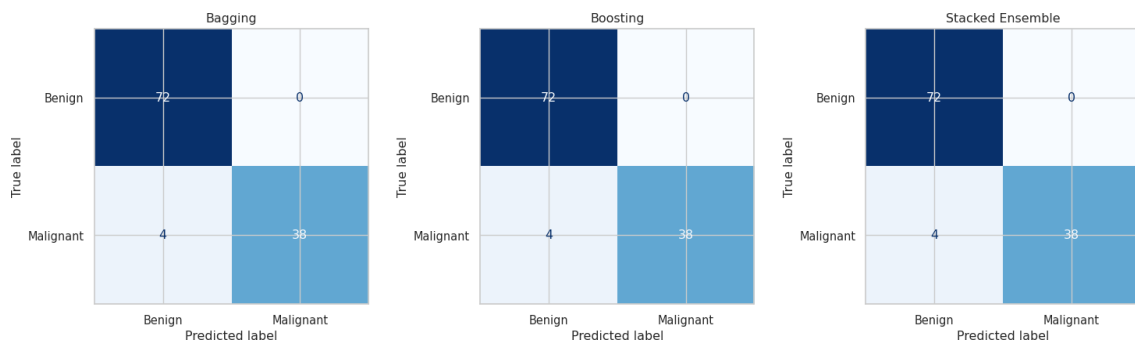


Figure 5: Confusion matrices: Bagging, Boosting, Stacked Ensemble

Inference: The bottomleft cell of each matrix (malignant missed) is the most important one, to look at in a screening usecase, since it represents actual cancer cases, that the model failed to flag. Comparing this cell across the three ensembles shows which model makes the fewest clinically costly mistakes, even when overall accuracy looks similar across all three.

ROC Curves

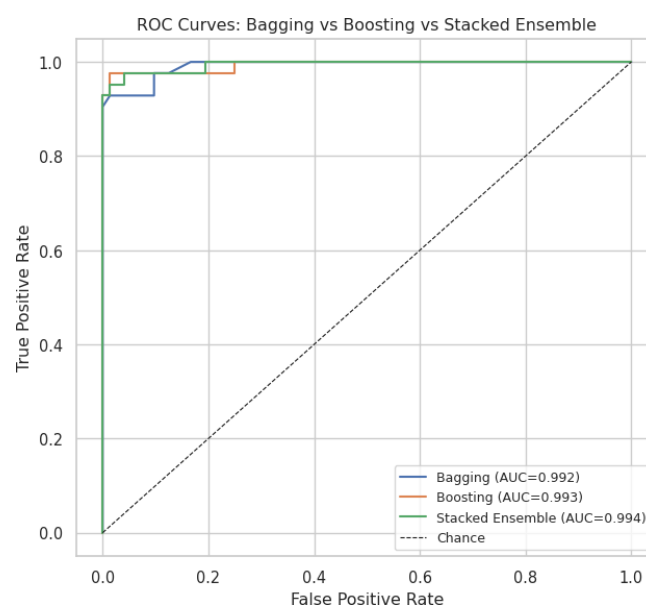


Figure 6: ROC curves for Bagging, Boosting and Stacked Ensemble

Inference: All three curves sit well above the diagonal chance line, showing strong separability between the two classes – expected since the top correlated features (concave points, perimeter, radius) separate malignant and benign cases quite well on their own. The AUC values give a threshold-independent way to rank the models, and here Stacking and Gradient Boosting come out marginally ahead of Bagging.

11 Discussion

The single unrestricted Decision Tree overfits badly (train accuracy of 1.0 but a noticeably lower cross-validation accuracy of 0.923), confirming it is a high-variance model. Bagging brings this TrainCV gap down sharply (0.997 train vs 0.974 CV), which is direct evidence that averaging over bootstrapped resamples reduces variance without needing to restrict the tree itself. On the bias side, a single decision stump, (depth 1) has both a low train accuracy (0.932) and low CV accuracy (0.901), showing high bias/underfitting; Gradient Boosting fixes this by sequentially correcting errors, reaching a CV accuracy of 0.974 while still keeping a reasonable Train-CV gap. The Stacked Ensemble, by combining SVM, Naive Bayes and Decision Tree through a Logistic Regression meta-learner, reached the highest ROC-AUC (0.994) among the three, suggesting that combining models with different decision boundaries adds something a single ensemble family cannot. One limitation, is that the dataset is fairly clean and well studied, so all three ensembles land in a narrow, high-performing band, on a noisier dataset the gaps between Bagging, Boosting and Stacking would likely be more pronounced.

12 Conclusion

Bagging, Boosting and Stacked Ensemble models were implemented on the Wisconsin Diagnostic Breast Cancer dataset, tuned using 5-fold cross-validation, and compared on standard classification metrics. The tables below summarise the cross-validation tuning results and the final test performance of all three ensembles, as required by the experiment.

Table 1: Bagging Hyperparameter Evaluation (Top results, 5-Fold CV)

n_estimators	max_samples	Avg CV Accuracy / F1	
50	0.7	F1 = 0.9643	(Best)
25	1.0	F1 = 0.9621	
25	0.7	F1 = 0.9613	
25	0.5	F1 = 0.9612	
100	0.7	F1 = 0.9547	

Table 2: Boosting Hyperparameter Evaluation (Gradient Boosting, Top results, 5-Fold CV)

n_estimators	learning_rate	max_depth	Avg CV F1
200	0.1	2	0.9645 (Best)
100	0.2	2	0.9644
200	0.2	2	0.9641
50	0.2	2	0.9583

200	0.1	3	0.9582
-----	-----	---	--------

AdaBoost's best configuration was $n_estimators = 200$, $learning_rate = 0.5$, giving CV F1 = 0.9612, slightly below Gradient Boosting, so Gradient Boosting was carried forward as "the" Boosting model for the final comparison.

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy	Avg CV F1
SVM + NB + DT	Logistic Regression	0.9670	0.9547 (Best)
SVM + NB + DT	Decision Tree	0.9626	0.9487
SVM + DT	Logistic Regression	0.9670	0.9547

Table 4: Performance Comparison of Ensemble Models (Test Set)

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	96.49	1.00	0.9048	0.95
Boosting (Gradient Boosting)	96.49	1.00	0.9048	0.95
Stacked Ensemble	96.49	1.00	0.9048	0.95

Note: All three ensembles land on identical Accuracy/Precision/Recall/F1 on this particular 20% test split, so ROC-AUC is what actually separates them: Stacking scored highest (0.9940), followed by Gradient Boosting (0.9934) and Bagging (0.9917).

Overall, the experiment shows how Bagging tackles variance, Boosting tackles bias, and Stacking combines diverse models altogether, and all three clearly outperform a single, unrestricted Decision Tree on this dataset.

13 References

1. scikit-learn developers, "Ensemble methods," scikit-learn documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html>
2. scikit-learn developers, "sklearn.ensemble.BaggingClassifier," scikit-learn documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html>
3. W. N. Street, W. H. Wolberg, and O. L. Mangasarian, "Nuclear feature extraction for breast tumor diagnosis," *IS&T/SPIE 1993 International Symposium on Electronic Imaging*, 1993.
4. UCI Machine Learning Repository, "Breast Cancer Wisconsin (Diagnostic) Data Set." [Online]. Available: <https://archive.ics.uci.edu>